

ASSIGNMENT 3

SAMPLE SOLUTION

Suppose my Roll No. is 006, Then I will consider 6 in Roll_No Variable. Also replacing respective lines ~~Now~~ with while & do while loop as per given key.

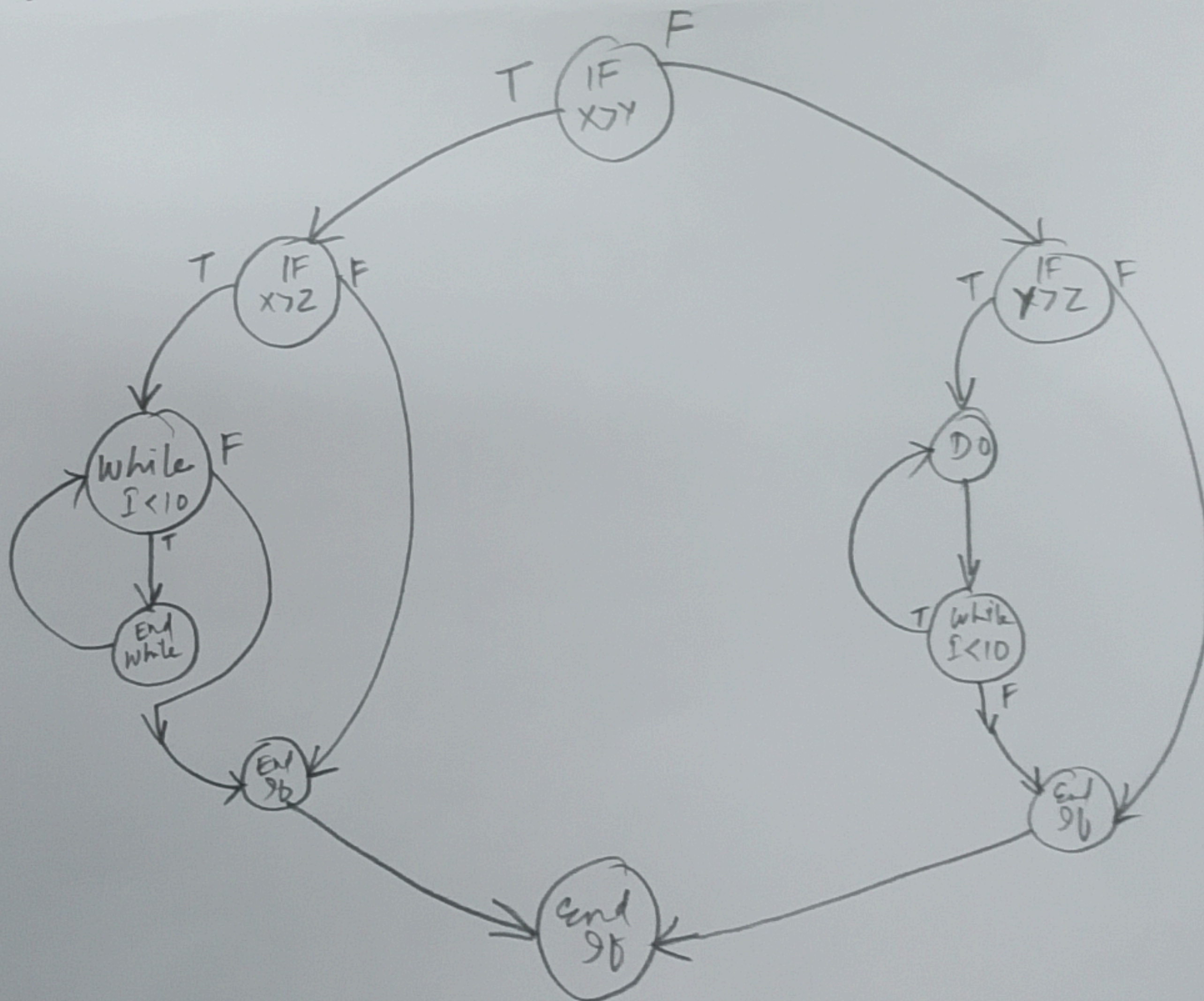
Step#1: Rewrite Code acc to the given Key.

```
1. READ Roll_No
2. A = Roll_No * -1
3. B = Roll_No * -10
4. C = Roll_No * 50
5. X = C + A - B
6. Y = C + B - A
7. Z = A + B + C
8. IF X > Y THEN
9.     IF X > Z THEN
10.        I = 0 → ②
11.        WHILE I < 10 DO
12.            PRINT A → ③
13.        END WHILE
14.        PRINT B → ④
15.    ELSE
16.        PRINT C → ⑤
17.    END IF
18. ELSE
19.     IF Y > Z THEN
20.        I = 0 → ⑥
21.        DO
22.            PRINT A → ⑦
23.        WHILE I < 10
24.        PRINT B → ⑧
25.    ELSE
26.        PRINT C → ⑨
27.    END IF
28. END IF
```

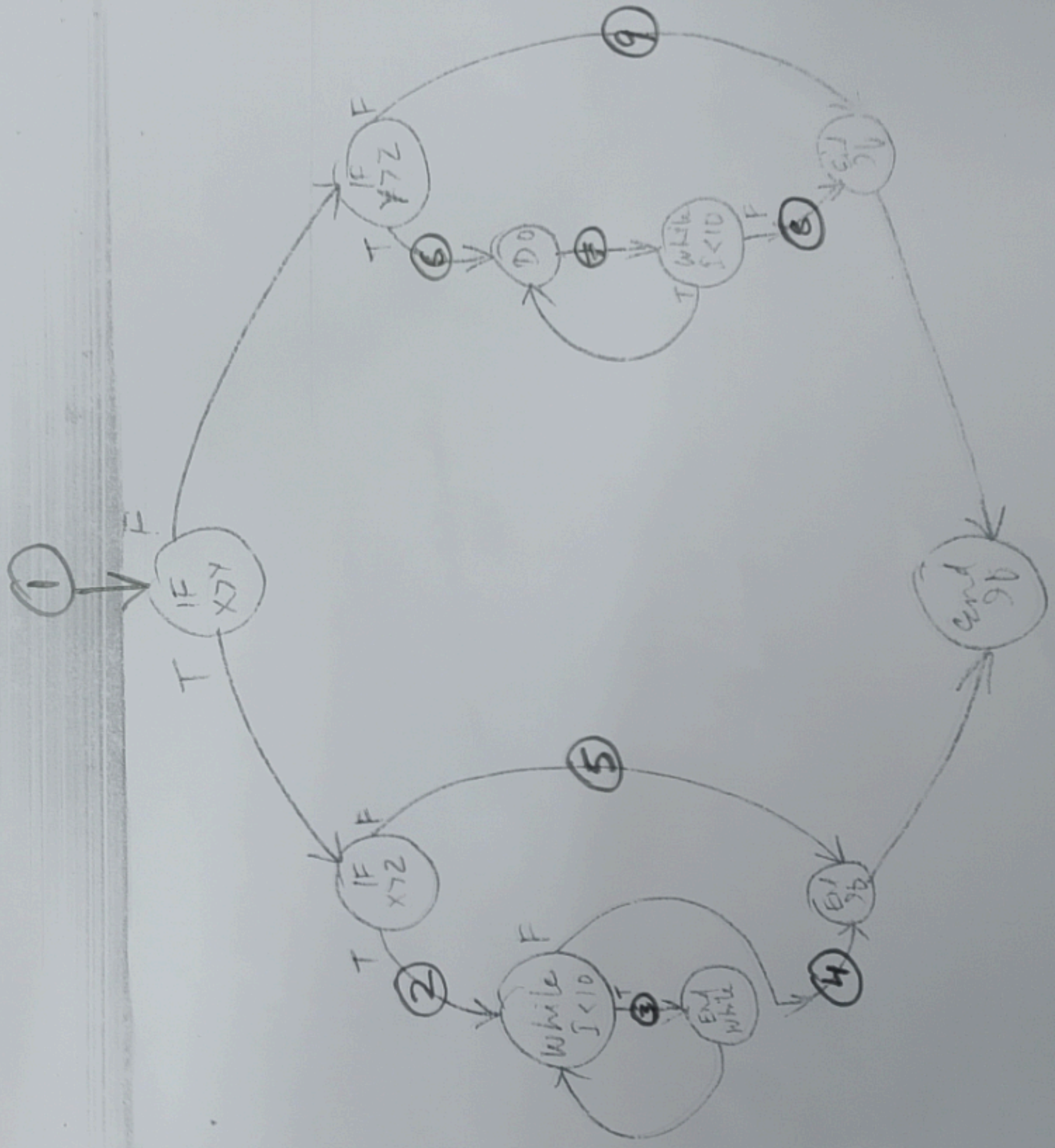

Sol (a) Control flow graph (CFG)

Consider only decision statements:

8. IF $X > Y$ THEN
9. IF $X > Z$ THEN
11. WHILE $I < 10$ DO
13. END WHILE
15. ELSE
17. END IF
18 ELSE
19. IF $Y > Z$ THEN
21. DO
23. WHILE $I < 10$
25. ELSE
27. END IF
28. END IF



c) hybrid flow graph (HFG).



Consider HFG & identify all paths with group of statements being executed. Ignore iteration path (upward arrow, don't consider it)

path 2: ① & ⑤

path 3: ①, ⑥, ⑦ & ⑧

path 4: ① & ⑨

Statements are unique to each path.
Therefore, we need to generate test data
paths. ~~for~~ so that max.

Therefore, we need 4 paths. ~~for~~ So that max. coverage could be achieved. Now, we have to select values of Roll No in such a way that all 4 paths could be executed.

Note: When we add Expected outcome with test data then test case

Test Data

Note: When we add Expected outcome with test data then it becomes a test case

	Roll_No	A	B	C	X	Y	Z	path executed
Same	6	-6	-60	300	354	246	234	path 1 (TT)
	500	-500	-5000	25000	29500	20500	19500	path 1 (TT)
Same	0	0	0	0	0	0	0	path 4 (FF)
	-10000	10000	100000	500000	4910000	5090000	5110000	path 4 (FF)
Rest of the 2 Cases are not possible.								

Sol d

Since there are total 4 paths, therefore we need 4 data set. but as only 2 are possible there 2 could be generated.

Solution is same as (c) means same test data is valid for max. decision Coverage for this question.

Sol (e)

4 test cases are required for 100% ~~test~~ statement Coverage but as only 2 cases are possible therefore, 100% statement Coverage is not possible.

Sol (f) Same answer as Sol (e).

... + data for max. statement

Sol
(18)

$$\text{Decision Coverage} = \frac{\text{Total No. of decision outcomes executed (paths)}}{\text{Total No. of decision outcomes (paths)}} \times 100$$

for Roll No 6, path 1 is executed,

$$\Rightarrow DC = \frac{1}{4} \times 100 = 25\%$$

Sol (18)

$$\text{Statement Coverage} = \frac{\text{Total No. of statements executed}}{\text{Total No. of statements}} \times 100$$

Total No. of statements = 28.

Roll No	A	B	C	X	Y	Z	line no. of statements executed	Total No. of statements executed	Statement Coverage.
6	-6	-60	300	354	246	234	1 to 15, 17, 28	17	17/28 x 100 = 60.71%